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DEPARTMENT OF INFORMATION TECHNOLOGY
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A Synopsis On
CipherDocs: A Secure AI-Powered Solution to Prevent Forgery of
Critical and Official Documents Using Polygon Blockchain-Based
Encryption and Verification.

By Project ID: B18

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1. PROBLEM STATEMENT

The widespread use of digital documents in education, governance, healthcare, and institutional processes has increased concerns related to document forgery, unauthorized modification, privacy leakage, and inefficient verification. Most existing document management and verification systems are centralized, making them vulnerable to single points of failure, lack of transparency, data tampering, and manipulation. Manual verification processes further increase time, operational cost, and the possibility of human error, limiting scalability in real-world scenarios such as academic credential, legal document, and official certificate validation.

While blockchain technology provides immutability and transparency, many existing blockchain-based solutions rely on networks with high transaction costs and limited scalability, making large-scale deployment impractical. Additionally, several systems focus only on hash-based integrity verification and lack intelligent AI-based validation for detecting forged or manipulated documents. Existing approaches also provide limited support for complete document lifecycle management, including automated revocation, expiry handling, adaptive security, and real-time verification. Hence, there is a need for a secure, scalable, and cost-efficient decentralized document issuance and verification system that integrates Polygon blockchain, IPFS-based storage, AES-256 encryption, SHA-256 hashing, adaptive key management, and Vision Language Model (VLM)-based AI validation to ensure authenticity, transparency, privacy preservation, and real-time document verification.

2. ABSTRACT

Significant challenges exist in ensuring the security of electronic (digital) documents, including forgery, unauthorized alterations, privacy leakage, inefficient verification mechanisms, and reduced public trust in conventional centralized document storage systems. This paper focuses on CipherDocs, its platform architecture, and the methodology adopted for the secure issuance and verification of documents. CipherDocs leverages Polygon blockchain technology and smart

contracts to establish a robust framework that preserves the integrity, authenticity, and transparent lifecycle management of critical records by permanently storing cryptographic hash values and essential document metadata on the blockchain. To address scalability and storage limitations, the platform utilizes decentralized off-chain storage through IPFS along with AES-256 encryption and adaptive key management for secure document protection. Additionally, the system integrates Vision Language Model (VLM)-based AI validation and contextual forgery detection to identify manipulated, forged, or semantically inconsistent documents beyond traditional hash-based verification.

CipherDocs also incorporates dynamic QR codes for instant document verification, real-time authentication, automated expiration and revocation policies, and AI-assisted trust evaluation mechanisms. By combining blockchain immutability, decentralized storage, adaptive security, AI-powered validation, and automated lifecycle management, CipherDocs provides a low-cost, scalable, secure, and intelligent solution for modern document verification systems. The proposed framework significantly enhances trust, transparency, privacy preservation, and fraud resistance, making it suitable for future digital governance, educational, legal, and institutional ecosystems.

3. MOTIVATION

The motivation behind choosing CipherDocs arises from the increasing number of document forgery cases, unauthorized alterations, AI-generated fake documents, and data breaches in digital document management systems used in critical sectors such as governance, education, healthcare, and legal compliance. Existing centralized systems fail to provide long-term trust, transparency, security, and intelligent validation due to single points of failure, limited control mechanisms, and dependence on manual verification processes. This project aims to address these real-world issues by offering a secure, decentralized, and tamper-resistant solution that ensures document integrity, authenticity, privacy preservation, and contextual verification using Polygon blockchain, IPFS storage, AES-256 encryption, adaptive security mechanisms, and Vision Language Model (VLM)-based AI validation.

CipherDocs is designed to contribute to ongoing research in blockchain-based verification by integrating cost-efficient blockchain infrastructure with AI-powered forgery detection, automated verification, dynamic QR authentication, and privacy-preserving techniques such as Zero-Knowledge Proofs. The system intends to reduce manual intervention, verification delays, and unnecessary exposure of sensitive information while maintaining scalability, usability, and real-time accessibility. The potential beneficiaries include government authorities, educational institutions, healthcare organizations, legal agencies, employers, and third-party verifiers who require reliable document validation. Ultimately, the motivation is to build a practical, intelligent, and user-friendly system that strengthens public trust in digital documents and protects society from evolving forgery threats in the digital era.

4. SDG TARGETED

The CipherDocs project directly supports selected **United Nations Sustainable Development Goals (SDGs)** by addressing challenges related to trust, transparency, security, and accessibility of digital documents. By leveraging blockchain-based verification, decentralized storage, and automated validation mechanisms, the project contributes to building reliable digital infrastructures and strengthening institutional processes.

SDG 4: Quality Education

CipherDocs contributes to **SDG 4 (Quality Education)** by enabling secure issuance and instant verification of academic credentials such as degrees, diplomas, transcripts, and certificates. The tamper-proof nature of blockchain-based document anchoring ensures long-term integrity and authenticity of educational records. Dynamic QR-code-based verification allows institutions and employers to validate credentials without manual intervention, reducing the prevalence of fake certificates and verification delays. This enhances trust in educational qualifications and supports fair access to academic and professional opportunities.

SDG 9: Industry, Innovation, and Infrastructure

The project aligns with **SDG 9 (Industry, Innovation, and Infrastructure)** by demonstrating an innovative and scalable digital infrastructure for document verification. CipherDocs leverages the Polygon blockchain to achieve low transaction costs and high throughput, making decentralized verification feasible for large-scale institutional use. The integration of decentralized storage and smart contracts showcases how modern digital infrastructure can be designed to be secure, resilient, and cost-efficient, supporting innovation in digital governance and enterprise systems.

SDG 16: Peace, Justice, and Strong Institutions

CipherDocs significantly supports **SDG 16 (Peace, Justice, and Strong Institutions)** by promoting transparency, accountability, and trust in document-related processes. Immutable blockchain audit records provide verifiable histories of document issuance, verification, revocation, and expiry, strengthening institutional accountability. By reducing reliance on centralized authorities and manual verification, the system minimizes opportunities for fraud, manipulation, and misuse of official documents. This contributes to more transparent and reliable institutional practices in governance, education, and administration.

5. LITERATURE REVIEW

5.1 Introduction of Literature survey

The literature survey presents a comprehensive review of existing blockchain-based document and certificate verification systems, highlighting the methodologies, technologies, and security mechanisms currently in use. Prior research predominantly focuses on leveraging public blockchains such as Ethereum and Polygon, smart contracts, IPFS, NFTs, and QR-code-based verification to ensure immutability, transparency, and trust in digital credentials. These approaches successfully address issues of data tampering and unauthorized modifications while establishing decentralized verification frameworks for academic and official documents.

However, the survey also reveals several limitations in existing systems, including high transaction costs, scalability challenges, poor user experience, limited automation, region-specific applicability, and insufficient privacy controls. Most solutions rely heavily on manual verification processes and static mechanisms, lacking intelligent forgery detection and real-time validation capabilities. These gaps indicate a clear scope for improvement by integrating cost-efficient blockchains, automated workflows, AI-based forgery detection, dynamic verification mechanisms, and privacy-preserving techniques. Addressing these limitations forms the foundation for developing advanced systems like CipherDocs that enhance security, usability, scalability, and trust.

Sr. No.	Title of the Paper	Publisher	Year	Methodology	Conclusion
1	Certifier dApp – Decentralized and Secured Certification System Using Blockchain	IEEE	2023	Ethereum blockchain with smart contracts and MetaMask for certificate issuance and verification	Ensures immutable and secure certification records but suffers from high gas fees and manual verification steps
2	A Comparative Study of Ethereum and Polygon for Implementing NFT-Based Certification Systems	IEEE	2024	Comparative implementation of NFT-based certification on Ethereum and Polygon Layer-2	Polygon provides faster transactions and lower costs, but the system is limited to NFT-based certificates
3	A Systematic Literature Review on Blockchain Based Systems for Academic Certificate Verification	Elsevier	2023	Review and analysis of 34 blockchain-based verification studies	Highlights strong security focus but identifies poor usability and high implementation costs

4	Decentralized Certificate Issuance and Verification System Using Ethereum Blockchain Technology	Springer	2025	Smart contract-based end-to-end certificate issuance and verification	Demonstrates feasibility but performance degrades under high load due to gas costs
5	ShikkhaChain: A Blockchain-Powered Academic Credential Verification System for Bangladesh	IEEE	2025	Blockchain-based credential issuance tailored to national education system	Effective for regional use but lacks forgery detection and scalability considerations
6	Doccert: Nostrification, Document Verification and Authenticity Blockchain Solution	Springer	2023	Blockchain framework for document authentication and nostrification	Provides secure verification but is limited in scope and incurs high operational costs
7	Paperless Everything: A Systematic Literature Review for the Design of Blockchain-Based Document Management Systems	Elsevier	2026	Smart contracts with IPFS and cryptographic hashing	Improves document management security but lacks full lifecycle automation
8	Decentralized Document Storage and Retrieval Using Blockchain, IPFS, and QR Code Technology for Secure and Tamper-Proof Access	IEEE	2025	Blockchain-integrated IPFS storage with QR-code-based access	Ensures tamper-proof storage, but verification is basic due to static QR codes

9	Document Forgery Detection: A Comprehensive Review	Springer	2025	Review of traditional and ML-based document forgery detection techniques	ML methods show promise, but real-time and post-compression verification remain unresolved
10	Blockchain-Based Framework for Document Verification	IEEE	2022	Smart contracts for document validation with blockchain immutability	Guarantees integrity but lacks privacy controls and cost optimization

6. OBJECTIVES AND SCOPE OF THE PROJECT

The primary objective of the CipherDocs project is to address the limitations of centralized document verification systems by designing a secure, decentralized, intelligent, and tamper-proof document issuance and verification platform using blockchain technology and AI-assisted validation. The specific objectives of the project are:

- To analyze the limitations of centralized document verification systems and address them using a decentralized blockchain-based approach
- To design a verification workflow that maintains document integrity, authenticity, and transparency throughout its lifecycle
- To investigate the effectiveness of Vision Language Model (VLM)-based AI techniques in detecting forged, manipulated, or semantically inconsistent digital documents
- To develop a privacy-aware and adaptive security mechanism using AES-256 encryption, dynamic key management, and secure third-party verification
- To evaluate the scalability, verification performance, and cost efficiency of the proposed system in real-world deployment scenarios using the Polygon blockchain

The scope of the CipherDocs project includes the development of a decentralized platform for issuing, storing, analyzing, and verifying digital documents using the Polygon blockchain and IPFS-based decentralized storage. The system focuses on preventing document forgery by

ensuring immutability, transparency, traceability, and AI-assisted contextual verification through blockchain-anchored cryptographic hashes, smart contract-based lifecycle management, and Vision Language Model (VLM)-based document analysis. Automated verification using QR codes, hash comparison, AI validation, and trust-based verification mechanisms enables secure real-time authentication without reliance on centralized authorities. CipherDocs is designed to support issuers, document holders, and third-party verifiers through secure wallet-based access and privacy-preserving mechanisms. However, the scope of this project is limited to digital document issuance, verification, and AI-assisted authenticity analysis, and does not include physical document handling, legal enforcement, or fully autonomous AI decision-making systems.

7. METHODOLOGY

The methodology adopted for the CipherDocs project is formulated based on a systematic analysis of existing literature on blockchain-based document verification, AI-assisted forgery detection, and decentralized trust systems. The literature review highlights that traditional centralized document management systems suffer from single points of failure, lack of transparency, high susceptibility to forgery, inefficient manual verification processes, and limited intelligent validation capabilities. While early blockchain-based solutions address immutability, many rely on high-cost public blockchains such as Ethereum, limiting scalability and real-world adoption. Recent studies further indicate that Layer-2 blockchain solutions such as Polygon provide significantly lower transaction costs and higher throughput, while Vision Language Models (VLMs) and AI-based semantic analysis improve the detection of manipulated, forged, or contextually inconsistent documents beyond traditional hash comparison techniques.

Based on these findings, the proposed methodology integrates blockchain immutability, cryptographic hashing, adaptive encryption, decentralized storage, AI-powered validation, and automated verification workflows. The system decentralizes trust, minimizes manual intervention, and ensures long-term integrity, transparency, and contextual authenticity of digital documents. CipherDocs utilizes SHA-256 hashing for integrity verification, AES-256 encryption

with adaptive key management for document security, Polygon blockchain smart contracts for lifecycle management, and IPFS for decentralized off-chain storage. Additionally, the methodology incorporates Vision Language Model (VLM)-based AI validation for semantic and visual forgery detection, along with dynamic QR-based real-time verification and trust-aware authentication mechanisms. The overall workflow replicates real-world document issuance and verification scenarios involving issuers, users, and third-party verifiers operating within a decentralized and AI-assisted blockchain ecosystem.

7.1 System Architecture

The CipherDocs system follows a modular, decentralized, and AI-assisted architecture that separates document issuance, storage, validation, and verification to eliminate single points of failure and improve scalability. The architecture consists of three primary entities: Issuer, User, and Verifier, all interacting through smart contracts deployed on the Polygon blockchain along with AI-powered verification modules and decentralized storage services.

7.1.1 Architectural Overview

During the issuance phase, the issuer uploads a digital document to the system. A cryptographic hash of the document is generated using SHA-256 hashing, while the document itself is encrypted using AES-256 encryption with adaptive key management policies. The generated hash, along with essential metadata such as document ID, issuance timestamp, expiry details, issuer information, and revocation status, is immutably recorded on the Polygon blockchain using smart contracts.

The actual encrypted document file is stored off-chain using decentralized storage mechanisms such as IPFS. This hybrid storage model significantly reduces on-chain storage overhead while preserving document availability, confidentiality, and integrity. Smart contracts govern the complete document lifecycle, including issuance, verification, expiry management, revocation, and access validation. Dynamic QR codes linked to blockchain records are generated for each issued document to enable secure and real-time third-party verification.

The architecture further integrates a Vision Language Model (VLM)-based AI validation layer that performs semantic and visual document analysis for detecting manipulated, forged, or AI-generated fake documents. AI-based contextual verification enhances traditional blockchain integrity checks by analyzing document structure, layout consistency, field-level modifications, and semantic authenticity.

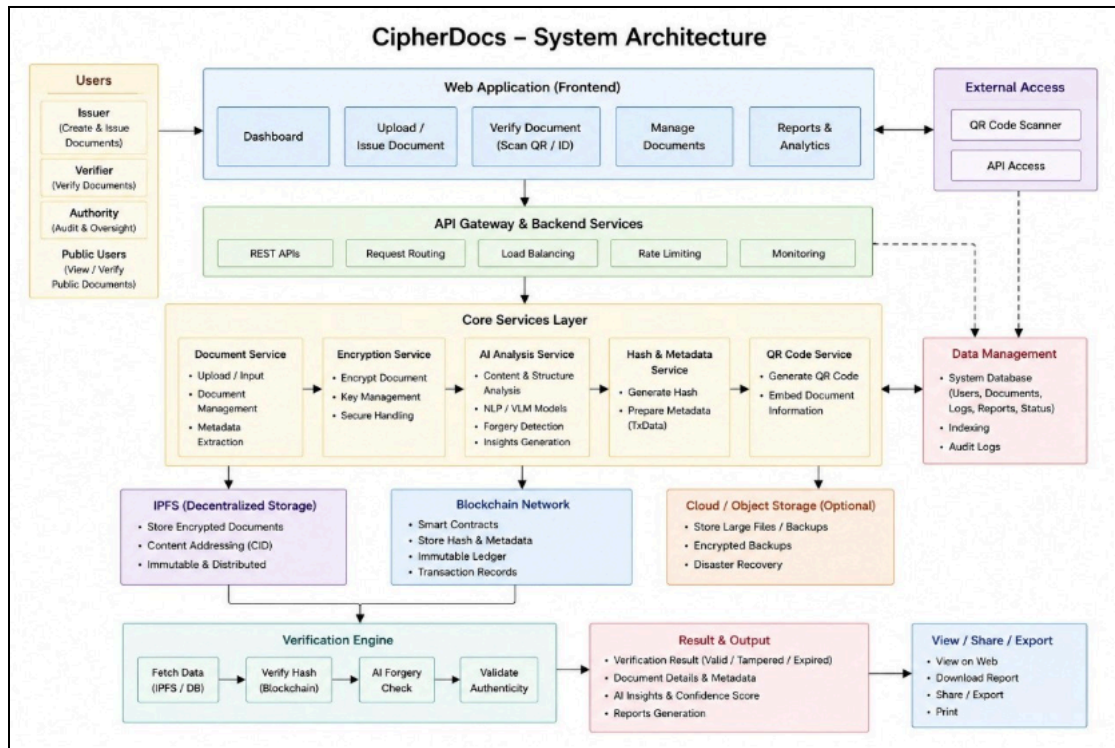


Figure 3.1 System Architecture

7.1.2 Issuer Module

The Issuer module initiates the trust chain by creating and issuing digital documents or certificates. Upon uploading a document, the system generates a SHA-256 cryptographic hash and encrypts the document using AES-256 encryption before storing it in IPFS. The issuer specifies validity parameters such as expiry date, document category, and revocation conditions, which are enforced automatically by blockchain smart contracts.

The module also integrates AI-assisted preprocessing and validation mechanisms for analyzing document structure and metadata consistency before issuance. Once issued, the document

becomes tamper-resistant, and any unauthorized modification attempts can be detected during verification through hash comparison and AI-based forgery analysis.

7.1.3 User Module

The User module allows legitimate document holders to securely access, download, manage, and share their digital documents. Wallet-based authentication mechanisms ensure that only authorized users can access sensitive records. Users can share documents with third-party verifiers through dynamic QR codes without exposing unnecessary personal information.

The system continuously monitors document validity, expiry status, and revocation conditions using blockchain smart contracts. AI-assisted document summaries and contextual verification responses can also be provided to improve usability for non-technical users.

7.1.4 Verifier Module

The Verifier module enables fast and reliable document authentication. Verification is initiated by scanning a QR code or entering a document identifier. The system retrieves the blockchain-anchored hash and compares it with the hash of the presented document.

Smart contracts are consulted to confirm document validity, expiry, and revocation status. Verification results are provided in near real-time without requiring manual checks or centralized authorities.

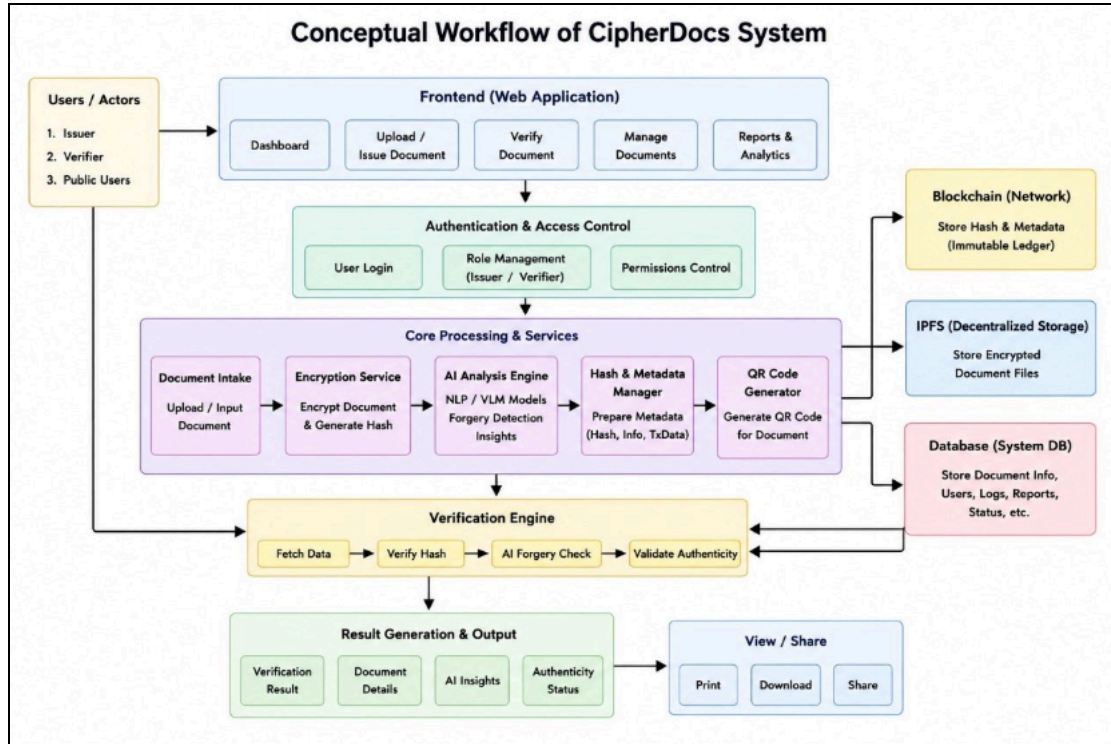


Figure 3.2 Depicts end-to-end flow of certificate issuance, storage, verification, and revocation.

7.2 Workflow and Methodology Flow

The overall workflow of CipherDocs is divided into four major phases: document issuance, AI-assisted validation, verification, and lifecycle management.

7.2.1 Document Issuance Flow

- Issuer uploads a digital document or certificate.
- System preprocesses the document and performs metadata validation.
- SHA-256 cryptographic hash of the document is generated.
- AES-256 encryption with adaptive key management is applied.
- Encrypted document is stored off-chain using IPFS.
- Hash and metadata are stored on the Polygon blockchain using smart contracts.
- Dynamic QR code is generated and linked with the blockchain verification record.

7.2.2 AI-Assisted Validation Flow

- Uploaded document is analyzed using Vision Language Models (VLMs).
- Visual and semantic features are extracted from the document.
- AI validation detects manipulated fields, layout inconsistencies, font mismatches, or suspicious modifications.
- Contextual analysis and trust-aware validation are performed using document metadata and issuer information.
- AI-generated trust assessment is combined with blockchain verification results.

7.2.3 Verification Flow

- Verifier scans the QR code or enters document ID.
- The system retrieves blockchain-anchored hash and metadata.
- Hash of the presented document is recomputed.
- Hash values are compared to detect tampering.
- Smart contracts validate document status, ownership, expiry, and revocation conditions.
- AI-based forgery analysis performs semantic and visual authenticity checks.
- Verification results are displayed instantly as Valid, Invalid, Revoked, Expired, or Suspicious.

7.2.4 Revocation and Expiry Management

Smart contracts automatically handle document revocation and expiry management. When a document is revoked or expires, its status is updated on the blockchain and associated QR verification becomes invalid. Adaptive security mechanisms also allow dynamic encryption policies, key rotation schedules, and trust-aware access management based on document sensitivity and issuer behavior. This prevents misuse of invalid or compromised documents without requiring manual intervention.

8. SOFTWARE REQUIREMENTS / SPECIFICATIONS

- **Blockchain Platform:** Polygon blockchain for immutable storage of document hashes, metadata, verification records, and lifecycle events with low transaction costs, fast confirmation time, and high scalability.
- **Smart Contracts:** Solidity-based smart contracts to manage document issuance, verification, revocation, expiry handling, trust validation, and secure lifecycle management of digital documents.
- **Decentralized Storage:** IPFS for secure off-chain storage of encrypted document files, while maintaining lightweight cryptographic references and metadata on the blockchain to reduce storage overhead.
- **Backend Framework:** Node.js and Express.js for handling APIs, blockchain interactions, cryptographic hashing, adaptive encryption workflows, QR code generation, AI service integration, and verification logic.
- **Frontend Interface:** Web-based user interface developed using modern web technologies such as Next.js, HTML, CSS, and JavaScript for issuers, users, and third-party verifiers.
- **Authentication Mechanism:** Wallet-based authentication using MetaMask or institutional Web3 wallets for secure user access, decentralized identity verification, and transaction authorization.
- **Verification Mechanism:** Dynamic QR code-based verification combined with SHA-256 cryptographic hash comparison, blockchain validation, document lifecycle checks, and AI-assisted authenticity analysis.
- **AI Validation Layer:** Vision Language Model (VLM)-based AI validation system for semantic analysis, contextual verification, forgery detection, layout consistency analysis, and suspicious document identification.
- **Cryptographic Security:** AES-256 encryption with adaptive key management and SHA-256 hashing for ensuring document confidentiality, integrity, and tamper resistance.
- **Blockchain Integration Libraries:** Web3.js or Ethers.js for interacting with Polygon smart contracts, blockchain transactions, and decentralized verification services.

- **AI and NLP Frameworks:** Integration of Vision Language Models (VLMs), OCR processing, and NLP-based contextual analysis for intelligent document validation and trust scoring.
- **Development Tools:** Hardhat or Truffle for smart contract development, testing, deployment, and blockchain debugging.

9. CONTRIBUTION OF THE PROJECT

The CipherDocs project makes a focused and meaningful contribution in the areas of Quality Education, Effective Governance, and Cyber Security, where document authenticity, integrity, and verifiability are critical. The project introduces a decentralized and AI-assisted document issuance and verification framework that prevents forgery by anchoring cryptographic hashes and metadata on the Polygon blockchain while securely storing encrypted documents off-chain using IPFS-based decentralized storage. The system further strengthens verification through AES-256 encryption, SHA-256 hashing, adaptive security mechanisms, and Vision Language Model (VLM)-based AI validation for detecting forged, manipulated, or contextually inconsistent documents. By automating verification and reducing dependence on centralized authorities, CipherDocs addresses practical challenges such as fake credentials, AI-generated forged documents, slow verification processes, lack of transparency, privacy concerns, and single points of failure in traditional document management systems. The project also contributes through dynamic QR-based real-time verification, automated revocation and expiry management, scalable low-cost blockchain deployment using Polygon, and intelligent forgery detection mechanisms, making the system more secure, transparent, scalable, and suitable for modern institutional and digital governance ecosystems.

9.1 Quality Education & Effective Governance

CipherDocs contributes directly to quality education and effective governance by enabling secure issuance and instant verification of academic and official documents. Educational certificates, degrees, transcripts, licenses, and government-issued records can be issued in a tamper-proof manner, ensuring long-term authenticity and trust. Each document is cryptographically hashed and immutably recorded on the blockchain, allowing verifiers to validate credentials at any time without depending on issuing authorities or manual checks.

The use of dynamic QR codes significantly improves verification efficiency by allowing institutions, employers, and government agencies to instantly validate documents. Smart contracts manage the document lifecycle, including revocation and expiry-based access control, ensuring that invalid or outdated documents cannot be misused. Immutable audit records provide transparency and traceability of all document-related activities, strengthening accountability in educational institutions and governance systems. As a result, CipherDocs helps reduce fake certificates, improves administrative efficiency, and enhances trust in public and academic credentials.

9.2 Communication, Networking, and Cyber Security

In the domain of communication, networking, and cyber security, CipherDocs strengthens digital trust by providing secure document access, controlled sharing, and verifiable authenticity across distributed networks. Wallet-based authentication ensures that only authorized issuers, users, and verifiers can access or share documents, reducing identity fraud and unauthorized access. The decentralized architecture eliminates single points of failure, improving resilience against cyber attacks and centralized data breaches.

Documents are stored off-chain using decentralized storage such as IPFS, while blockchain records maintain cryptographic references and verification history. This design ensures secure availability of documents while minimizing exposure of sensitive data. Immutable blockchain audit logs record verification activities, enabling monitoring, accountability, and forensic analysis when required. Through these mechanisms, CipherDocs contributes to secure digital communication systems where document integrity, traceability, and trust are essential.

10. EXPECTED OUTCOMES

The CipherDocs project is expected to contribute a secure, scalable, intelligent, and cost-efficient framework for issuing and verifying critical and official documents using blockchain technology and AI-assisted validation. By leveraging the Polygon blockchain, the system demonstrates how low transaction costs, high throughput, and fast confirmation times can make decentralized document verification practical for real-world institutional deployment. The project contributes a tamper-resistant mechanism for document integrity by anchoring SHA-256 cryptographic hashes and metadata on the blockchain, ensuring that any unauthorized modification to issued documents can be reliably detected. Additionally, the integration of AES-256 encryption with adaptive key management strengthens document confidentiality and security throughout the document lifecycle.

Another key contribution of this project is the integration of decentralized storage, blockchain-based verification, and Vision Language Model (VLM)-based AI validation. By storing encrypted documents off-chain using IPFS and maintaining only essential metadata on-chain, CipherDocs provides a balanced approach that reduces storage overhead while preserving transparency, availability, and fault tolerance. The AI-assisted validation layer enhances traditional hash-based verification by performing semantic and visual forgery detection, contextual analysis, layout consistency checks, and trust-aware verification. This design addresses scalability and intelligent validation limitations commonly observed in traditional blockchain-only document management systems.

The project also contributes to improving verification efficiency, transparency, and reliability by enabling automated QR-code-based real-time document validation without dependence on centralized authorities or manual verification processes. Smart contracts support automated revocation, expiry handling, and lifecycle management, while immutable blockchain audit trails enhance accountability and trust among issuers, users, and verifiers. Overall, CipherDocs demonstrates a practical and deployable model for secure digital document management, offering valuable architectural insights and reusable methodologies for applications in education, governance, healthcare, legal systems, and enterprise environments.

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